



# Signals and Systems

## Tutorial 3-2

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## Problem 3.11, 3.30, 3.37, 3.49

# Reviews

- **Periodicity Properties of DT Complex Exponentials;**
- **DT Fourier Series Representation;**
- **DT Fourier Series Pair;**
- **DTFS of Periodic Square Wave;**
- **Properties of DT Fourier Series;**
- **Frequency response of LTI system;**
- **Filtering**

# Reviews



## CTFS pair

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t},$$
$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_T x(t) e^{-jk(2\pi/T)t} dt.$$

## DTFS pair

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n},$$
$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}.$$

# Reviews

## CTFS properties

TABLE 3.1 PROPERTIES OF CONTINUOUS-TIME FOURIER SERIES

| Property                                                              | Section | Periodic Signal                                                                                                                      | Fourier Series Coefficients                                                                                                                                |
|-----------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                       |         | $x(t)$ } Periodic with period $T$ and<br>$y(t)$ } fundamental frequency $\omega_0 = 2\pi/T$                                          | $a_k$<br>$b_k$                                                                                                                                             |
| Linearity                                                             | 3.5.1   | $Ax(t) + By(t)$                                                                                                                      | $Aa_k + Bb_k$                                                                                                                                              |
| Time Shifting                                                         | 3.5.2   | $x(t - t_0)$                                                                                                                         | $a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$                                                                                                         |
| Frequency Shifting                                                    |         | $e^{jM\omega_0 t} = e^{jM(2\pi/T)t} x(t)$                                                                                            | $a_{k-M}$                                                                                                                                                  |
| Conjugation                                                           | 3.5.6   | $x^*(t)$                                                                                                                             | $a_{-k}^*$                                                                                                                                                 |
| Time Reversal                                                         | 3.5.3   | $x(-t)$                                                                                                                              | $a_{-k}$                                                                                                                                                   |
| Time Scaling                                                          | 3.5.4   | $x(\alpha t), \alpha > 0$ (periodic with period $T/\alpha$ )                                                                         | $a_k$                                                                                                                                                      |
| Periodic Convolution                                                  |         | $\int_T x(\tau)y(t - \tau)d\tau$                                                                                                     | $Ta_k b_k$                                                                                                                                                 |
| Multiplication                                                        | 3.5.5   | $x(t)y(t)$                                                                                                                           | $\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$                                                                                                                   |
| Differentiation                                                       |         | $\frac{dx(t)}{dt}$                                                                                                                   | $jk\omega_0 a_k = jk \frac{2\pi}{T} a_k$                                                                                                                   |
| Integration                                                           |         | $\int_{-\infty}^t x(\tau) d\tau$ (finite valued and periodic only if $a_0 = 0$ )                                                     | $\left(\frac{1}{jk\omega_0}\right)a_k = \left(\frac{1}{jk(2\pi/T)}\right)a_k$                                                                              |
| Conjugate Symmetry for Real Signals                                   | 3.5.6   | $x(t)$ real                                                                                                                          | $\begin{cases} a_k = a_{-k}^* \\ \Re\{a_k\} = \Re\{a_{-k}\} \\ \Im\{a_k\} = -\Im\{a_{-k}\} \\  a_k  =  a_{-k}  \\ \angle a_k = -\angle a_{-k} \end{cases}$ |
| Real and Even Signals                                                 | 3.5.6   | $x(t)$ real and even                                                                                                                 | $a_k$ real and even                                                                                                                                        |
| Real and Odd Signals                                                  | 3.5.6   | $x(t)$ real and odd                                                                                                                  | $a_k$ purely imaginary and odd                                                                                                                             |
| Even-Odd Decomposition of Real Signals                                |         | $\begin{cases} x_e(t) = \mathcal{E}\{x(t)\} & [x(t) \text{ real}] \\ x_o(t) = \mathcal{O}\{x(t)\} & [x(t) \text{ real}] \end{cases}$ | $\begin{cases} \Re\{a_k\} \\ \Im\{a_k\} \end{cases}$                                                                                                       |
| Parseval's Relation for Periodic Signals                              |         |                                                                                                                                      |                                                                                                                                                            |
| $\frac{1}{T} \int_T  x(t) ^2 dt = \sum_{k=-\infty}^{+\infty}  a_k ^2$ |         |                                                                                                                                      |                                                                                                                                                            |

## DTFS properties

TABLE 3.2 PROPERTIES OF DISCRETE-TIME FOURIER SERIES

| Property                                                                               | Periodic Signal                                                                                                                                                               | Fourier Series Coefficients                                                                                                                                |
|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                        | $x[n]$ } Periodic with period $N$ and<br>$y[n]$ } fundamental frequency $\omega_0 = 2\pi/N$                                                                                   | $a_k$ } Periodic with<br>$b_k$ } period $N$                                                                                                                |
| Linearity                                                                              | $Ax[n] + By[n]$                                                                                                                                                               | $Aa_k + Bb_k$                                                                                                                                              |
| Time Shifting                                                                          | $x[n - n_0]$                                                                                                                                                                  | $a_k e^{-jk(2\pi/N)n_0}$                                                                                                                                   |
| Frequency Shifting                                                                     | $e^{jM(2\pi/N)n} x[n]$                                                                                                                                                        | $a_{k-M}$                                                                                                                                                  |
| Conjugation                                                                            | $x^*[n]$                                                                                                                                                                      | $a_{-k}^*$                                                                                                                                                 |
| Time Reversal                                                                          | $x[-n]$                                                                                                                                                                       | $a_{-k}$                                                                                                                                                   |
| Time Scaling                                                                           | $x_{(m)}[n] = \begin{cases} x[n/m], & \text{if } n \text{ is a multiple of } m \\ 0, & \text{if } n \text{ is not a multiple of } m \end{cases}$ (periodic with period $mN$ ) | $\frac{1}{m} a_k$ (viewed as periodic with period $mN$ )                                                                                                   |
| Periodic Convolution                                                                   | $\sum_{r=-\infty}^{+\infty} x[r]y[n - r]$                                                                                                                                     | $Na_k b_k$                                                                                                                                                 |
| Multiplication                                                                         | $x[n]y[n]$                                                                                                                                                                    | $\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$                                                                                                                   |
| First Difference                                                                       | $x[n] - x[n - 1]$                                                                                                                                                             | $(1 - e^{-jk(2\pi/N)})a_k$                                                                                                                                 |
| Running Sum                                                                            | $\sum_{k=-\infty}^n x[k]$ (finite valued and periodic only if $a_0 = 0$ )                                                                                                     | $\left(\frac{1}{(1 - e^{-jk(2\pi/N)})}\right)a_k$                                                                                                          |
| Conjugate Symmetry for Real Signals                                                    | $x[n]$ real                                                                                                                                                                   | $\begin{cases} a_k = a_{-k}^* \\ \Re\{a_k\} = \Re\{a_{-k}\} \\ \Im\{a_k\} = -\Im\{a_{-k}\} \\  a_k  =  a_{-k}  \\ \angle a_k = -\angle a_{-k} \end{cases}$ |
| Real and Even Signals                                                                  | $x[n]$ real and even                                                                                                                                                          | $a_k$ real and even                                                                                                                                        |
| Real and Odd Signals                                                                   | $x[n]$ real and odd                                                                                                                                                           | $a_k$ purely imaginary and odd                                                                                                                             |
| Even-Odd Decomposition of Real Signals                                                 | $\begin{cases} x_e[n] = \mathcal{E}\{x[n]\} & [x[n] \text{ real}] \\ x_o[n] = \mathcal{O}\{x[n]\} & [x[n] \text{ real}] \end{cases}$                                          | $\begin{cases} \Re\{a_k\} \\ \Im\{a_k\} \end{cases}$                                                                                                       |
| Parseval's Relation for Periodic Signals                                               |                                                                                                                                                                               |                                                                                                                                                            |
| $\frac{1}{N} \sum_{n=-\infty}^{+\infty}  x[n] ^2 = \sum_{k=-\infty}^{+\infty}  a_k ^2$ |                                                                                                                                                                               |                                                                                                                                                            |

# Reviews



## CTFS pair

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t},$$

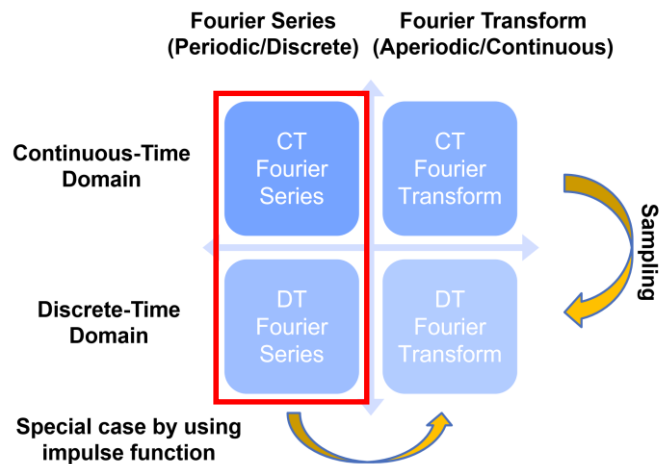
$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_T x(t) e^{-jk(2\pi/T)t} dt.$$

## DTFS pair

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n},$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}.$$

## Overview on Frequency Analysis



**TABLE 5.3** SUMMARY OF FOURIER SERIES AND TRANSFORM EXPRESSIONS

|                   | Continuous time                                                                                                           |                                                                                                                  | Discrete time                                                                                                  |                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|                   | Time domain                                                                                                               | Frequency domain                                                                                                 | Time domain                                                                                                    | Frequency domain                                                                                                    |
| Fourier Series    | $x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$<br>continuous time<br>periodic in time                           | $a_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$<br>discrete frequency<br>aperiodic in frequency       | $x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$<br>discrete time<br>periodic in time                   | $a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}$<br>discrete frequency<br>periodic in frequency |
| Fourier Transform | $x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$<br>continuous time<br>aperiodic in time | $X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$<br>continuous frequency<br>aperiodic in frequency | $x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$<br>discrete time<br>aperiodic in time | $X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n}$<br>continuous frequency<br>periodic in frequency  |

“The duality property for discrete-time Fourier series implies that the Fourier series coefficients for the periodic sequence  $a_k$  are the values of  $(1/N)x[-n]$  (i.e., are proportional to the values of the original signal, reversed in time).”

## Problem 3.11

**3.11.** Suppose we are given the following information about a signal  $x[n]$ :

1.  $x[n]$  is a real and even signal.
2.  $x[n]$  has period  $N = 10$  and Fourier coefficients  $a_k$ .
3.  $a_{11} = 5$ .
4.  $\frac{1}{10} \sum_{n=0}^9 |x[n]|^2 = 50$ .

Show that  $x[n] = A \cos(Bn + C)$ , and specify numerical values for the constants  $A$ ,  $B$ , and  $C$ .

## Solution 3.11

**3.11** Since the Fourier series coefficients repeat every  $N=10$ , we have  $a_1 = a_{11} = 5$ . Furthermore, since  $x[n]$  is real and even,  $a_k$  is also real and even. Therefore  $a_1 = a_{-1} = 5$ . We are also given that

$$\frac{1}{10} \sum_{n=0}^9 |x[n]|^2 = 50$$

Using Parseval's relation,

$$\sum_{k=-\infty}^{\infty} |a_k|^2 = 50$$

$$\sum_{k=-1}^8 |a_k|^2 = 50$$

$$|a_{-1}|^2 + |a_1|^2 + a_0^2 + \sum_{k=2}^8 |a_k|^2 = 50$$

$$a_0^2 + \sum_{k=2}^8 |a_k|^2 = 0$$

## Solution 3.11

Therefore  $a_k = 0$  for  $k=2, \dots, 8$ , Now using the synthesis eq. (3.94), we have

$$\begin{aligned} x[n] &= \sum_{k=\langle N \rangle} a_k e^{j\frac{2\pi}{N}kn} = \sum_{k=-1}^8 a_k e^{j\frac{2\pi}{10}kn} \\ &= 5e^{j\frac{2\pi}{10}n} + 5e^{-j\frac{2\pi}{10}n} = 10 \cos\left(\frac{\pi}{5}n\right) \end{aligned}$$

## Problem 3.30

**3.30.** Consider the following three discrete-time signals with a fundamental period of 6:

$$x[n] = 1 + \cos\left(\frac{2\pi}{6}n\right), \quad y[n] = \sin\left(\frac{2\pi}{6}n + \frac{\pi}{4}\right), \quad z[n] = x[n]y[n].$$

- (a) Determine the Fourier series coefficients of  $x[n]$ .
- (b) Determine the Fourier series coefficients of  $y[n]$ .
- (c) Use the results of parts (a) and (b), along with the multiplication property of the discrete-time Fourier series, to determine the Fourier series coefficients of  $z[n] = x[n]y[n]$ .
- (d) Determine the Fourier series coefficients of  $z[n]$  through direct evaluation, and compare your result with that of part (c).

## DT Fourier Series Pair $\left(\omega_o = \frac{2\pi}{N}\right)$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_o n} = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n},$$
$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_o n} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}.$$

## Solution 3.30

3.30.(a) The nonzero FS coefficients of  $x(t)$  are  $a_0=1, a_1=a_{-1}=1/2$

(b) The nonzero FS coefficient FS coefficient of  $x(t)$  are  $b_1 = b_{-1}^* = e^{-j\pi/4} / 2$

(c) Using the multiplication property, we know that

$$z[n] = x[n]y[n] \xleftrightarrow{FS} c_k = \sum_{l=-2}^2 a_l b_{k-l}$$

This implies that the nonzero Fourier series coefficients of  $z[n]$  are  $c_0 = \cos(\pi/4)/2$ ,  
 $c_1 = c_{-1}^* = e^{-j\pi/4} / 2, c_2 = c_{-2}^* = e^{-j\pi/4} / 4$

(d) We have

$$\begin{aligned} z[n] &= \sin\left(\frac{2\pi}{n} + \frac{\pi}{4}\right) + \sin\left(\frac{2\pi}{6}n + \frac{\pi}{4}\right) \cos\left(\frac{2\pi}{6}n\right) \\ &= \sin\left(\frac{2\pi}{6}n + \frac{\pi}{4}\right) + \frac{1}{2} \left[ \sin\left(\frac{4\pi}{6}n + \frac{\pi}{4}\right) + \sin\left(\frac{\pi}{4}\right) \right] \\ &= \dots \quad (\text{expand } \sin(x+y) \text{ and use Euler's formula}) \\ &= \dots \quad (\text{in form of } \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}) \end{aligned}$$

Euler's formula:

$$e^{jx} = \cos x + j \sin x$$

$$\sin \phi = \frac{1}{2j} (e^{j\phi} - e^{-j\phi})$$

$$\cos \phi = \frac{1}{2} (e^{j\phi} + e^{-j\phi})$$

## Problem 3.37

3.37. Consider a discrete-time LTI system with impulse response

$$h[n] = \left(\frac{1}{2}\right)^{|n|}.$$

Find the Fourier series representation of the output  $y[n]$  for each of the following inputs:

(a)  $x[n] = \sum_{k=-\infty}^{\infty} \delta[n - 4k]$

(b)  $x[n]$  is periodic with period 6 and

$$x[n] = \begin{cases} 1, & n = 0, \pm 1 \\ 0, & n = \pm 2, \pm 3 \end{cases}$$

# DT Fourier Series Pair $\left(\omega_o = \frac{2\pi}{N}\right)$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_o n} = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n},$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_o n} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}.$$

$$H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} h[n] e^{-j\omega n}$$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_o n} \longrightarrow \boxed{h[n]} \longrightarrow y[n] = \sum_{k=-\infty}^{\infty} H(e^{jk\omega_o}) a_k e^{jk\omega_o n}$$

$$a_k \longrightarrow \underbrace{H(e^{jk\omega_o}) a_k}_{\text{"gain"}}$$

## Solution 3.37

3.37 The frequency response of the system may be easily shown to be

$$H(e^{j\omega}) = \frac{1}{1 - \frac{1}{2}e^{-j\omega}} - \frac{1}{1 - 2e^{-j\omega}}$$

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=-\infty}^{+\infty} h[n] e^{-j\omega n} \\ &= \sum_{n=-\infty}^{+\infty} \left(\frac{1}{2}\right)^{|n|} e^{-j\omega n} \\ &= \sum_{n=0}^{+\infty} \left(\frac{1}{2}\right)^n e^{-j\omega n} + \sum_{n=-\infty}^{-1} \left(\frac{1}{2}\right)^{-n} e^{-j\omega n} \\ &= \frac{1}{1 - \frac{1}{2}e^{-j\omega}} + \frac{\frac{1}{2}e^{j\omega}}{1 - \frac{1}{2}e^{j\omega}} \\ &= \frac{1}{1 - \frac{1}{2}e^{-j\omega}} - \frac{1}{1 - 2e^{j\omega}} \end{aligned}$$

## Solution 3.37

**3.37** The frequency response of the system may be easily shown to be

$$H(e^{j\omega}) = \frac{1}{1 - \frac{1}{2}e^{-j\omega}} - \frac{1}{1 - 2e^{-j\omega}}$$

(a) the Fourier series coefficients of  $x[n]$  are

$$a_k = 1/4, \text{ for all } k$$

Also,  $N=4$ . Therefore, the Fourier series coefficients of  $y[n]$  are

$$b_k = a_k H(e^{j2k\pi/N}) = \frac{1}{4} \left[ \frac{1}{1 - \frac{1}{2}e^{-j\pi k/2}} - \frac{1}{1 - 2e^{-j\pi k/2}} \right].$$

## Solution 3.37

**3.37** The frequency response of the system may be easily shown to be

$$H(e^{j\omega}) = \frac{1}{1 - \frac{1}{2}e^{-j\omega}} - \frac{1}{1 - 2e^{-j\omega}}$$

(b) In this case, the Fourier series coefficients of  $x[n]$  are

$$\begin{aligned} a_k &= \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk \frac{2\pi}{N} n} \\ &= \frac{1}{6} \sum_{n=-3}^2 x[n] e^{-jk \frac{\pi}{3} n} \\ &= \frac{1}{6} (e^{-jk \frac{\pi}{3} \times (-1)} + e^{-jk \frac{\pi}{3} \times 0} + e^{-jk \frac{\pi}{3} \times 1}) \\ &= \frac{1}{6} (e^{jk \frac{\pi}{3}} + 1 + e^{-jk \frac{\pi}{3}}) \\ &= \frac{1}{6} [1 + 2 \times \frac{1}{2} (e^{jk \frac{\pi}{3}} + e^{-jk \frac{\pi}{3}})] \\ &= \frac{1}{6} [1 + 2 \cos \frac{k\pi}{3}] \end{aligned}$$

Also  $N=6$ . Therefore, the Fourier series coefficients of  $y[n]$  are

$$b_k = a_k H(e^{j2k\pi/N}) = \frac{1}{6} [1 + 2 \cos(k\pi/3)] \left[ \frac{1}{1 - \frac{1}{2}e^{-j\pi k/3}} - \frac{1}{1 - 2e^{-j\pi k/3}} \right]$$

# Problem 3.49

**3.49.** Let  $x[n]$  be a periodic sequence with period  $N$  and Fourier series representation

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}. \quad (\text{P3.49-1})$$

(a) Suppose that  $N$  is even and that  $x[n]$  in eq. (P3.49-1) satisfies

$$x[n] = -x\left[n + \frac{N}{2}\right] \text{ for all } n.$$

Show that  $a_k = 0$  for all even integers  $k$ .

(b) Suppose that  $N$  is divisible by 4. Show that if

$$x[n] = -x\left[n + \frac{N}{4}\right] \text{ for all } n,$$

then  $a_k = 0$  for every value of  $k$  that is a multiple of 4.

(c) More generally, suppose that  $N$  is divisible by an integer  $M$ . Show that if

$$\sum_{r=0}^{(N/M)-1} x\left[n + r\frac{N}{M}\right] = 0 \text{ for all } n,$$

then  $a_k = 0$  for every value of  $k$  that is a multiple of  $M$ .

## Solution 3.49

3.49 (a) The FS coefficients are given by

$$\begin{aligned}
 a_k &= \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi n k}{N}} \\
 &= \frac{1}{N} \sum_{n=0}^{N/2-1} x[n] e^{-j \frac{2\pi n k}{N}} + \frac{1}{N} \sum_{n=N/2}^{N-1} x[n] e^{-j \frac{2\pi n k}{N}} \\
 &= \frac{1}{N} \sum_{n=0}^{N/2-1} x[n] e^{-j \frac{2\pi n k}{N}} - \frac{e^{-j \pi k}}{N} \sum_{n=0}^{(N/2)-1} x[n] e^{-j \frac{2\pi n k}{N}} \\
 &= 0,
 \end{aligned}$$

For k even

# Solution 3.49

3.49 (a) The FS coefficients are given by

$$\begin{aligned} a_k &= \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi n k}{N}} \\ &= \frac{1}{N} \sum_{n=0}^{N/2-1} x[n] e^{-j \frac{2\pi n k}{N}} + \frac{1}{N} \sum_{n=N/2}^{N-1} x[n] e^{-j \frac{2\pi n k}{N}} \\ &= \frac{1}{N} \sum_{n=0}^{N/2-1} x[n] e^{-j \frac{2\pi n k}{N}} - \frac{e^{-j \pi k}}{N} \sum_{n=0}^{(N/2)-1} x[n] e^{-j \frac{2\pi n k}{N}} \\ &= 0, \end{aligned}$$

For k even

(b) By adopting an approach similar to part (a), We may show that

$$\begin{aligned} a_k &= \frac{1}{N} \left[ \sum_{n=0}^{\frac{N}{4}-1} \{ 1 - e^{-j \pi k / 2} + e^{-j \pi k} - e^{-3 j \pi k / 2} \} x[n] e^{-j \frac{2\pi n k}{N}} \right] \\ &= 0 \\ \text{For } k=4r, \quad r \in R \end{aligned}$$

# Problem 3.49

**3.49.** Let  $x[n]$  be a periodic sequence with period  $N$  and Fourier series representation

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}. \quad (\text{P3.49-1})$$

(a) Suppose that  $N$  is even and that  $x[n]$  in eq. (P3.49-1) satisfies

$$x[n] = -x\left[n + \frac{N}{2}\right] \text{ for all } n.$$

Show that  $a_k = 0$  for all even integers  $k$ .

(b) Suppose that  $N$  is divisible by 4. Show that if

$$x[n] = -x\left[n + \frac{N}{4}\right] \text{ for all } n,$$

then  $a_k = 0$  for every value of  $k$  that is a multiple of 4.

(c) More generally, suppose that  $N$  is divisible by an integer  $M$ . Show that if

$$\sum_{r=0}^{M-1} x\left[n + r\frac{N}{M}\right] = 0 \text{ for all } n,$$

then  $a_k = 0$  for every value of  $k$  that is a multiple of  $M$ .

# Solution 3.49

3.49 (a) The FS coefficients are given by

$$\begin{aligned} a_k &= \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi n k}{N}} \\ &= \frac{1}{N} \sum_{n=0}^{N/2-1} x[n] e^{-j \frac{2\pi n k}{N}} + \frac{1}{N} \sum_{n=N/2}^{N-1} x[n] e^{-j \frac{2\pi n k}{N}} \\ &= \frac{1}{N} \sum_{n=0}^{N/2-1} x[n] e^{-j \frac{2\pi n k}{N}} - \frac{e^{-j \pi k}}{N} \sum_{n=0}^{(N/2)-1} x[n] e^{-j \frac{2\pi n k}{N}} \\ &= 0, \end{aligned}$$

For k even

(b) By adopting an approach similar to part (a), We may show that

$$\begin{aligned} a_k &= \frac{1}{N} \left[ \sum_{n=0}^{N-1} \{ 1 - e^{-j \pi k / 2} + e^{-j \pi k} - e^{-3 j \pi k / 2} \} x[n] e^{-j \frac{2\pi n k}{N}} \right] \\ &= 0 \end{aligned}$$

For  $k=4r, \quad r \in R$

(c) If  $N/M$  is an integer, We may generalize the approach of part (a) to show that

$$a_k = \frac{1}{N} \left[ \sum_{n=0}^{B-1} \{ 1 - e^{-j \pi} + e^{-j \pi 4r} + e^{-j \pi 2(M-1)r} \} x[n] e^{-j \frac{2\pi n k}{N}} \right]$$

Where  $B=N/M$  and  $r=k/M$ .

$a_k = 0, \quad \text{if } k \text{ is a multiple of } M.$